

Exercise 2.13. For each of the following pairs of numbers, list all of their common divisors (positive and negative!), and find $\gcd(a, b)$.

(a) $a = 12, b = 330$

(b) $a = -36, b = 64$

(c) $a = 7, b = -27$

Solution.

We need to list **all common divisors**, including both positive and negative divisors, and then find the greatest common divisor.

Recall that a number d is a common divisor of a and b if

$$d \mid a \quad \text{and} \quad d \mid b.$$

The gcd is taken to be **positive**.

(a) $a = 12, b = 330$

The positive divisors of 12 are

$$1, 2, 3, 4, 6, 12.$$

The positive divisors of 330 include

$$1, 2, 3, 5, 6, 10, 11, 15, 22, 30, 33, 55, 66, 110, 165, 330.$$

The common positive divisors are

$$1, 2, 3, 6.$$

Including the negative divisors, all common divisors are

$$\boxed{\pm 1, \pm 2, \pm 3, \pm 6}.$$

Therefore,

$$\boxed{\gcd(12, 330) = 6}.$$

(b) $a = -36$, $b = 64$

The divisors of -36 are the positive and negative divisors of 36 , while the divisors of 64 are the positive and negative divisors of 64 .

Since

$$36 = 2^2 \cdot 3^2, \quad 64 = 2^6,$$

their greatest common divisor is

$$2^2 = 4.$$

The common positive divisors are

$$1, 2, 4.$$

Including the negative divisors,

$$\boxed{\pm 1, \pm 2, \pm 4}.$$

Therefore,

$$\boxed{\gcd(-36, 64) = 4}.$$

(c) $a = 7$, $b = -27$

The positive divisors of 7 are

$$1, 7,$$

while the positive divisors of 27 are

$$1, 3, 9, 27.$$

The only common positive divisor is 1 . Including negative divisors, the common divisors are

$$\boxed{\pm 1}.$$

Therefore,

$$\boxed{\gcd(7, -27) = 1}.$$